

INT201 Decision, Computation and Language

Lecture 4 – Regular Language

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Definition

Previous: A language is regular if it is recognized by some **DFA** $\Rightarrow$ DFA $\Rightarrow$ regular

Now: A language is regular if and only if some **NFA** recognizes it. $NFA \Leftrightarrow$ regular

Some operations on languages: Union, Concatenation and Kleene star

$$S, a \in S \stackrel{\text{close}}{\Rightarrow} f(a) \in S$$

Closed under operation



A collection S of objects is **closed** under operation f if applying f to members of S always returns an object still in S .

Regular languages are indeed closed under the regular operations (e.g. union, concatenation, star ...)



Constructive proof.

Regular Languages Closed Under Union

The set of regular languages is closed under the union operation.

i.e. A and B are regular languages over the same alphabet Σ , then $A \cup B$ is also a regular language.

Proof:

- Since A and B are regular languages, there are finite automata $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ that accept A and B , respectively.
- In order to prove that $A \cup B$ is regular, we have to construct a finite automaton M that accepts $A \cup B$. In other words, M must have the property that for every string $w \in \Sigma^*$:



Regular Languages Closed Under Union

Proof

$M_1, M_2 \Rightarrow M$

Given M_1 and M_2 that $A = L(M_1)$ and $B = L(M_2)$, we can define $M = (Q, \Sigma, \delta, q, F)$:

- Let
- $Q = Q_1 \times Q_2 = \{(q_1, q_2): q_1 \in Q_1 \text{ and } q_2 \in Q_2\}$
 - Σ is same as the alphabet of A and B
 - $q = (q_1, q_2)$
 - $F = \{(q_1, q_2): q_1 \in F_1 \text{ or } q_2 \in F_2\}$
 - $\delta: Q \times \Sigma \rightarrow Q$

table

$$\delta((q_1, q_2), a) = (\delta(q_1, a), \delta(q_2, a)), a \in \Sigma$$

$\Rightarrow$ transition function



Regular Languages Closed Under Union

Proof

$$\text{2nd: } \Rightarrow \delta^*(q_1, q_2), w = \{ \delta^*(q_1, w), \delta^*(q_2, w) \}$$



$$\delta^*(q_1, w) \in F \text{ or } \delta^*(q_2, w) \in F_2$$

$$\Rightarrow M \text{ accepts } w \Leftrightarrow \delta^*(q_1, w) \in F_1 \text{ or } \delta^*(q_2, w) \in F_2$$

$$\Rightarrow M \text{ accepts } w \Leftrightarrow M_1 \text{ or } M_2 \text{ accepts } w$$



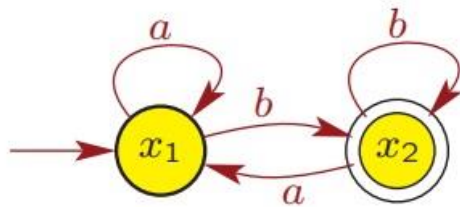
Example

Consider the following DFAs and languages over $\Sigma = \{a, b\}$:

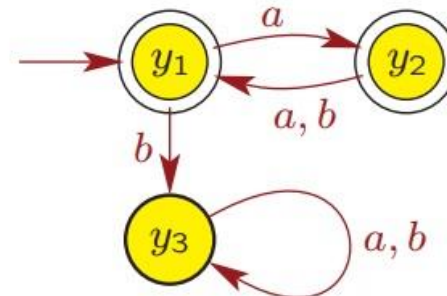
DFA M_1 recognizes $A_1 = L(M_1)$

DFA M_2 recognizes $A_2 = L(M_2)$

DFA M_1 for A_1



DFA M_2 for A_2



DFA M for $A_1 \cup A_2$?

$\delta_1(a, x_1)$
 $\delta_1(a, y_1)$

$\delta_2(a, x_1)$
 $\delta_2(b, x_1)$



Example

$$\textcircled{1} \quad Q: (Q_1 \times Q_2) \rightarrow \{(x_1, y_1), (x_1, y_2), (x_1, y_3), (x_2, y_1), (x_2, y_2), (x_2, y_3)\}.$$

$$q_1 = (x_1, x_2)$$

$$q_2 = (x_1, x_2, x_3)$$

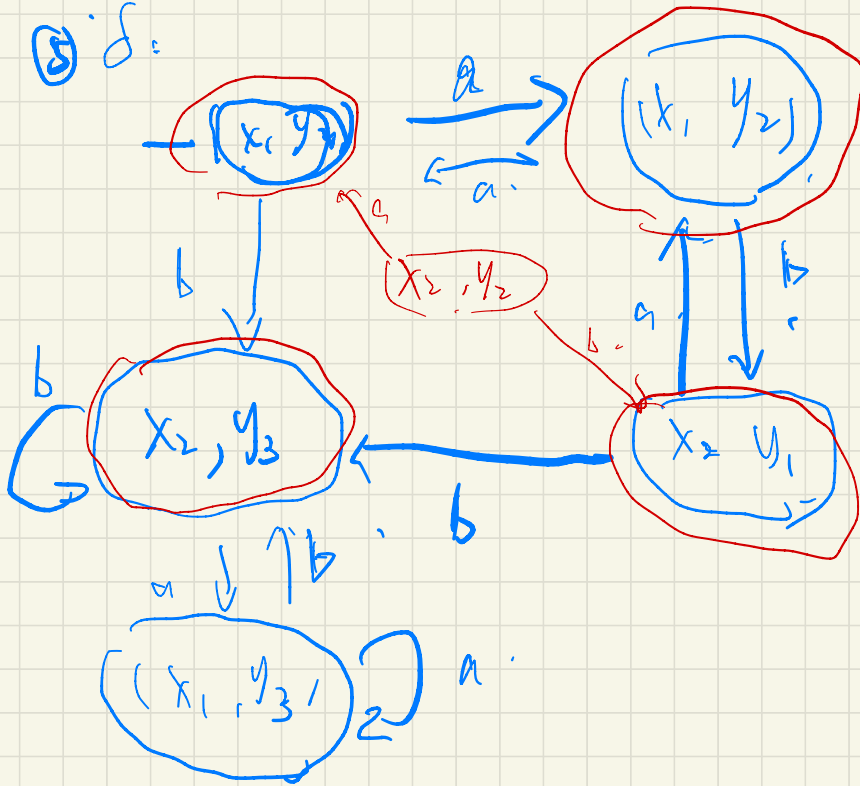
$$\textcircled{2} \quad \Sigma = \{a, b\}$$

$$\textcircled{3} \quad q = (x_1, y_1)$$

$$\textcircled{4} \quad T \equiv T_1 \times T_2 = \{(x_1, y_1), (x_1, y_2)\} \times \{x_2, y_1\} \cup \{(x_2, y_2)\} \times \{x_2, y_3\}$$



⑤ δ :



add ~~the~~ miss one $x_2 x_2$

circle accepting states.

Regular Languages Closed Under Union

How to prove this from the perspective of NFA?

way 2. construct δ tuple.

Proof

Consider the following NFAs:

NFA $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognizes $A_1 = L(M_1)$

NFA $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognizes $A_2 = L(M_2)$

We assume that $\underline{Q_1} \cap Q_2 = \emptyset$

We will construct an NFA $M = (Q, \Sigma, \delta, q, F)$



Regular Languages Closed Under Union

Proof

- $Q = \{q_0\} \cup Q_1 \cup Q_2$
 $\neq$
- q_0 is the start state of M

- $F = F_1 \cup F_2$

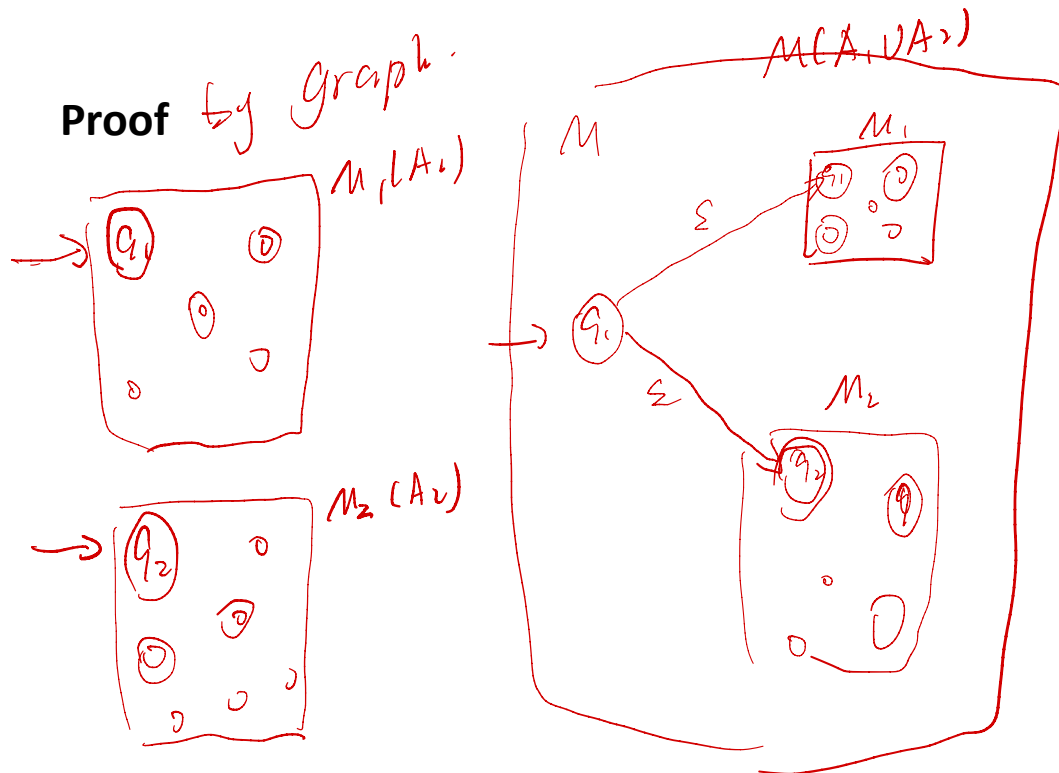
- $\delta: Q \times \Sigma_\epsilon \rightarrow P(Q)$ is defined as: For any $r \in Q$ and for any $a \in \Sigma_\epsilon$

$$\delta(r, a) = \begin{cases} \delta_1(r, a) & \text{if } r \in Q_1 \\ \delta_2(r, a) & \text{if } r \in Q_2 \\ \{q_1, q_2\} & \text{if } r = q_0 \text{ and } a = \epsilon \\ \emptyset & \text{if } r = q_0 \text{ and } a \neq \epsilon \end{cases}$$

$$\delta_1, \delta_2 \Rightarrow \delta$$



Regular Languages Closed Under Union



Regular Languages Closed Under Concatenation

The concatenation of A_1 and A_2 is defined as:

$$A_1 A_2 = \{ww' : w \in A_1 \text{ and } w' \in A_2\}$$

Proof

Consider the following NFAs:

NFA $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognizes $A_1 = L(M_1)$

NFA $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognizes $A_2 = L(M_2)$

We will construct an NFA $M = (Q, \Sigma, \delta, q, F)$ for $A_1 A_2$



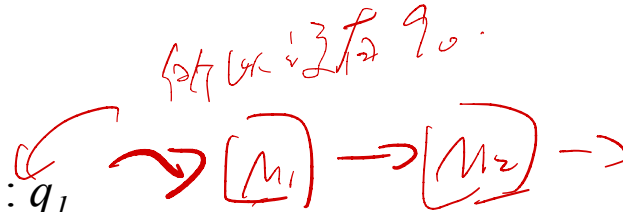
Regular Languages Closed Under Concatenation

5-tuple.

Proof

- $Q = Q_1 \cup Q_2$

- M has the same start state as $M_1: q_1$



- Set of accept states of M is same as $M_2: F_2$

- $\delta: Q \times \Sigma_\epsilon \rightarrow P(Q)$ is defined as: For any $r \in Q$ and for any $a \in \Sigma_\epsilon$

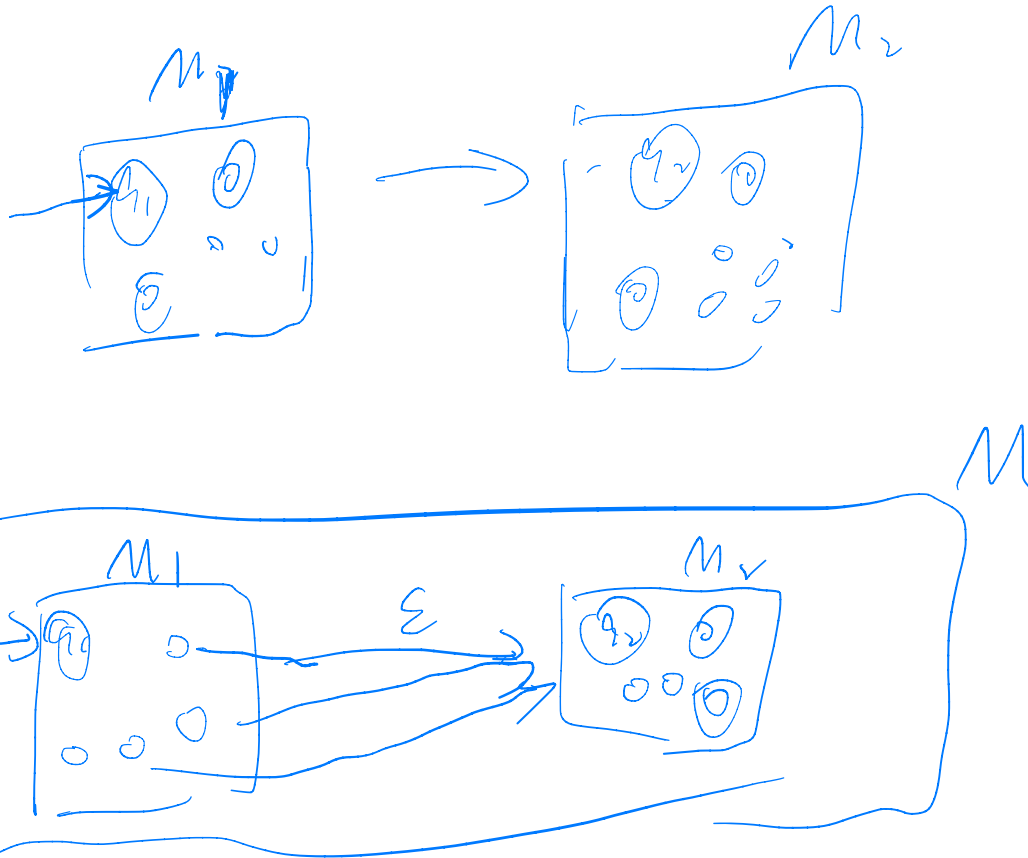
$$\delta(r, a) = \begin{cases} \delta_1(r, a) & \text{if } r \in Q_1 \text{ and } a \neq \epsilon \\ \delta_1(r, a) & \text{if } r \in F_1, \text{ and } a \neq \epsilon \\ \delta_1(r, a) \cup \{q_2\} & \text{if } r \in F_1, \text{ and } a = \epsilon \\ \delta_2(r, a) & \text{if } r \in Q_2. \end{cases}$$

how to connect M_1 with M_2 .



Regular Languages Closed Under Concatenation

Proof



Regular Languages Closed Under Kleene star

The star of A is defined as:

$$A^* = \{u_1 u_2 \dots u_k : k \geq 0 \text{ and } u_i \in A \text{ for all } i = 1, 2, \dots, k\}$$

Proof

Consider the following NFA:

NFA $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognizes $A = L(M_1)$

We will construct an NFA $M = (Q, \Sigma, \delta, q, F)$ for A^*



Regular Languages Closed Under Kleene star

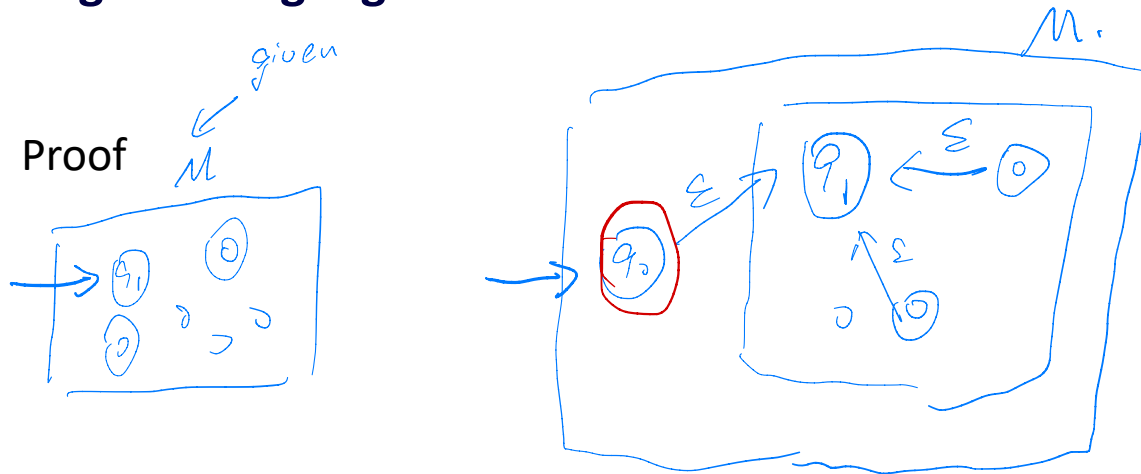
Proof

- $Q = \{q_0\} \cup Q_1$
- q_0 is the start state of M
- $F = \{q_0\} \cup F_1$
- $\delta : Q \times \Sigma_\epsilon \rightarrow P(Q)$ is defined as: For any $r \in Q$ and for any $a \in \Sigma_\epsilon$

$$\delta(r, a) = \begin{cases} \delta_1(r, a) & \text{if } r \in Q_1 \text{ and } r \notin F_1 \\ \delta_1(r, a) & \text{if } r \in F_1 \text{ and } a \neq \epsilon \\ \delta_1(r, a) \cup \{q_0\} & \text{if } r \in F_1, a = \epsilon \\ \{q_0\} & \text{if } r = q_0, a = \epsilon \\ \emptyset & \text{if } r = q_0, a \neq \epsilon \end{cases}$$



Regular Languages Closed Under Kleene star



Regular Languages Closed Under Complement and Intersection

证明证明.

The set of regular languages is closed under the complement and intersection operations:

- If A is a regular language over the alphabet Σ , then the complement:

$$\bar{A} = \{w \in \Sigma^* : w \notin A\}$$

is also a regular language.

- If A_1 and A_2 are regular languages over the same alphabet Σ , then the intersection:

$$A_1 \cap A_2 = \{w \in \Sigma^* : w \in A_1 \text{ and } w \in A_2\}$$

is also a regular language.



Regular Expressions

Regular expressions are means to describe certain languages.

Definition.

Example

Consider the expression:

$$(0 \cup 1)01^*$$

The language described by this expression is the set of all binary strings satisfy:

- that start with either 0 or 1 (this is indicated by $(0 \cup 1)$),
- for which the second symbol is 0 (this is indicated by 0),
- that end with zero or more 1s (this is indicated by 1^*).

*means =
 $\epsilon, 1, 11, \dots$ infinite 1s*



Example

The language $\{w : w \text{ contains exactly two 0s}\}$ is described by the expression:

$1^*01^*01^*$ $\Rightarrow$ 去一个?

The language $\{w : w \text{ contains at least two 0s}\}$ is described by the expression:

$(1^*0)^*0(1^*0)^*0(1^*0)^*$

The language $\{w : 1011 \text{ is a substring of } w\}$ is described by the expression:

$(1^*0)^*(01)^*(1^*0)^*$



Formal Definition of regular expressions

Let Σ be a non-empty alphabet.

1. ϵ is a regular expression.
2. $\emptyset$ is a regular expression.
3. For each $a \in \Sigma$, a is a regular expression.
4. If R_1 and R_2 are regular expressions, then $R_1 \cup R_2$ is a regular expression.
5. If R_1 and R_2 are regular expressions, then $R_1 R_2$ is a regular expression.
6. If R is a regular expression, then R^* is a regular expression.



Example

Given $(0 \cup 1)^* 101 (0 \cup 1)^*$, prove it is a regular expression (note: $\Sigma = \{0, 1\}$).


just follow definition



Formal Definition of regular expressions

If R is a regular expression, then $L(R)$ is the **language** generated (or described or defined) by R .

Let Σ be a non-empty alphabet.

1. The regular expression ϵ describes the language $\{\epsilon\}$.
2. The regular expression $\emptyset$ describes the language $\emptyset$.
3. For each $a \in \Sigma$, the regular expression a describes the language $\{a\}$.
4. Let R_1 and R_2 be regular expressions and let L_1 and L_2 be the languages described by them, respectively. The regular expression $R_1 \cup R_2$ describes the language $L_1 \cup L_2$.
5. Let R_1 and R_2 be regular expressions and let L_1 and L_2 be the languages described by them, respectively. The regular expression $R_1 R_2$ describes the language $L_1 L_2$.
6. Let R be a regular expression and let L be the language described by it. The regular expression R^* describes the language L^* .



Example

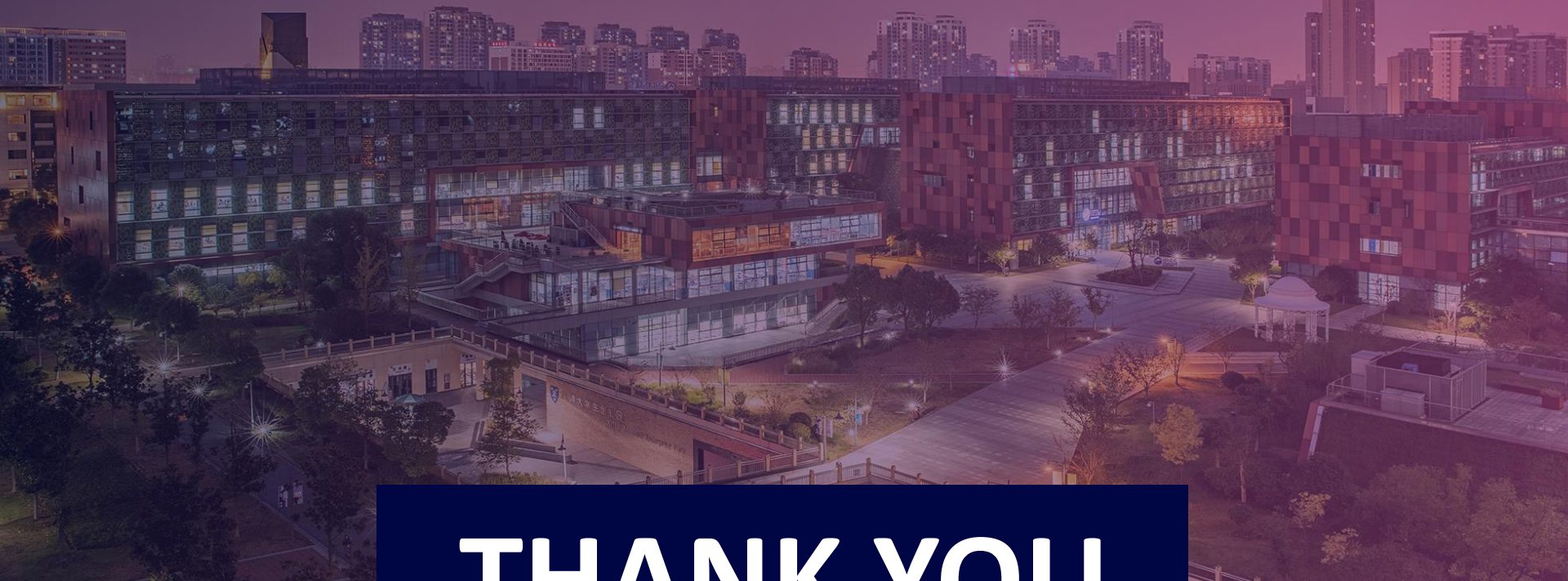
$(0 \vee 1)^* 1$

Given a regular expression $(0 \vee 1)^* 1$, it describes the language:

$\{ \epsilon, 01, 011, 01011, \dots, 1, 11, \dots \}$

equal $0^* 1^*$?





THANK YOU



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